

AFIAS BNP

INTENDED USE

AFIAS BNP is a fluorescence immunoassay (FIA) for the quantitative determination of cardiac BNP (B-type Natriuretic Peptide) level in human whole blood/plasma. It is useful as an aid in management and monitoring of Heart Failure(HF).
For *in vitro* diagnostic use only.

INTRODUCTION

B-type natriuretic peptide (BNP) is a cardiac neurohormone that derives from the precursor pre-proBNP, containing 134 amino acids and including a signal peptide of 26 amino acids. proBNP, produced by cleavage of the signal peptide, is further split into BNP, which is considered to be the biologically active hormone, and an inactive amino terminal fragment, NT-proBNP. Blood measurements of BNP have been shown to be useful in differentiating dyspnoea caused by congestive heart failure (CHF) from dyspnoea related to other causes.
AFIAS BNP Assay quantitatively measures the concentration of BNP, providing a physician with an accurate tool to assess prognosis in patients with chronic heart failure.

PRINCIPLE

The test uses a sandwich immunodetection method.
The detector antibodies in buffer bind to antigens in the sample, forming antigen-antibody complexes, and migrate onto nitrocellulose matrix to be captured by the other immobilized streptavidin on a test strip.
More antigens in the sample will form more antigen-antibody complexes which lead to stronger fluorescence signal by detector antibodies, which is processed by the instrument for AFIAS tests to show BNP concentration in the sample.

COMPONENTS

- AFIAS BNP** consists of ‘cartridges’.
- Each sealed aluminum pouch contains two cartridges.
 - Each cartridge packaged in an aluminum pouch has three components including a cartridge part, a detector part and a diluent part.
 - The cartridge part contains the membrane called a test strip which has streptavidin at the test line, and chicken IgY at the control line.
 - The detector part has 2 granules containing anti-BNP-fluorescence conjugate, biotin-anti-BNP conjugate, anti-chicken IgY-fluorescence conjugate, and sodium azide as a preservative in Tris buffer.
 - The diluent part contains tween 20 in sodium phosphate buffer.

WARNINGS AND PRECAUTIONS

- For *in vitro* diagnostic use only.
- Follow instructions and procedures described in this ‘Instructions for use’.
- Use only fresh samples and avoid direct sunlight.
- Lot numbers of all the test components (cartridge and ID chip) must match each other.
- Do not interchange the test components between different lots or use the test components after the expiration date, either of which might yield incorrect test result(s).
- Do not reuse cartridges, c-tips, or tips. A cartridge should be used for testing one sample only.
- The cartridge should remain sealed in its original pouch until just before use. Do not use a cartridge, if the pouch is damaged or has already been opened.
- Frozen sample should be thawed only once. For shipping, samples must be packed in accordance with local regulations. Sample with severe hemolysis and/or hyperlipidemia must not be used.
- If test components and/or sample are stored in refrigerator, then allow cartridge and sample to be at room temperature for approximately 30 minutes before use.
- The instrument for AFIAS tests may generate slight vibration during use.
- Used cartridges, C-tips and pipette tips should be handled carefully and discarded by an appropriate method in accordance with relevant local regulations.
- The cartridge contains sodium azide (NaN₃), and it may cause certain health issues like convulsions, low blood pressure, low heart rate, loss of consciousness, lung injury and respiratory failure. Avoid contact with skin, eyes, and clothing. In case of contact, rinse immediately with running water.
- No Biotin interference was observed in **AFIAS BNP** when biotin concentration in the sample was below 5 ng/ml. If a patient has been taking biotin at dosage of more than 0.03 mg a day, it is recommended to test again 24 hours after discontinuation of biotin intake.
- **AFIAS BNP** will provide accurate and reliable results subject to the below conditions.
 - **AFIAS BNP** should be used only in conjunction with the instrument for AFIAS tests.
 - Have to use recommended anticoagulant.

Recommended anticoagulant
K ₃ EDTA

- **C-tip should be used when the following conditions are met.**
 - C-tip provided with the kit is recommended to obtain correct test result.
 - Whole blood should be immediately tested after collection.
 - Do not perform a test with C-tip on General Mode. It might cause an erroneous result.
 - Excess whole blood around the C-tip should be wiped off.
 - In order to avoid cross-contamination, please do not re-use C-tip for multiple samples.
 - AFIAS cartridge should be inserted and positioned in the cartridge holder prior to the blood sample collection
 - While collecting blood, be careful not to create air

bubbles in the C-tip.

LIMITATIONS OF THE TEST SYSTEM

- The test may yield false positive result(s) due to the cross-reactions and/or non-specific adhesion of certain sample components to the capture/detector antibodies.
- The test may yield false negative result(s) due to the non-responsiveness of the antigens to the antibodies which is the most common if the epitope is masked by some unknown components, so therefore not being able to be detected or captured by the antibodies. The instability or degradation of the antigens with time and/or temperature may also cause false negative result as it makes the antigens unrecognizable by the antibodies.
- Other factors may interfere with the test and cause erroneous results, such as technical/procedural errors, degradation of the test components/reagents or presence of interfering substances in the test samples.
- Any clinical diagnosis based on the test result must be supported by a comprehensive judgment of the concerned physician in conjunction with clinical symptoms and other relevant test results.

STORAGE AND STABILITY

Storage condition			
Component	Storage Temperature	Shelf life	Note
Cartridge	2 – 30°C	20 months	Unopened
	2 – 30°C	1 month	Resealed

- Return an unused cartridge to the spare cartridge zipper bag containing the desiccant pack. Reseal along entire edge of zip-seal.

MATERIALS SUPPLIED

REF SMFP-83	
Components of AFIAS BNP	
■ Cartridge box:	
- Cartridge	24
- Pipette tip (zipper bag)	24
- Spare cartridge zipper bag	1
- ID chip	1
- Instructions for use	1

MATERIALS REQUIRED BUT SUPPLIED ON DEMAND

- Following items can be purchased separately with **AFIAS BNP**. Please contact our sales division for more information.
- Instrument for AFIAS tests
 - **AFIAS-1**
 - **AFIAS-3**
 - **AFIAS-6**
 - **AFIAS-10**
 - **C-tip (Zipper bag, 30 µL)**
 - **Boditech BNP Control**
 - **Boditech BNP Calibrator**

REF	FPRR019
REF	FPRR040
REF	FPRR020
REF	FPRR038
REF	CFPO-199
REF	CFPO-304
REF	CFPO-305

SAMPLE COLLECTION AND PROCESSING

- The sample type for **AFIAS BNP** is human whole blood/plasma.
- It is recommended to test the sample within 8 hours at room temperature and within 24 hours at 2-8 °C after collection.
 - The sample (plasma) should be separated from the clot by centrifugation within 3 hours after the collection of whole blood.
 - The samples (whole blood, plasma) may be stored for 24 hours at 2-8 °C prior to being tested. If testing will be delayed more than 24 hours, sample (plasma) should be frozen at -20 °C.
 - The sample (plasma) stored frozen at -20 °C for 12 months showed no performance difference.
 - However, the whole blood sample should not be kept in a freezer in any case.
 - As a repeated freeze-thaw cycle may affect the test result, do not refreeze previously frozen samples.
 - Collection of whole blood sample using C-tip
 - ① Hold the C-tip horizontally and touch the surface of the blood with the tip of the C-tip.
 - ② Capillary action will automatically draw the blood sample to C-tip and stop.
 - ③ Wipe off any excess blood around the tip.
 - ④ Double-check if whole blood is filled accurately in the C-tip and the instrument for AFIAS tests is ready for a test on the ‘C-tip mode’.

TEST SETUP

- Check the components of the **AFIAS BNP** as described below. : Cartridges, pipette tips, C-tips, an ID chip, a spare cartridge zipper bag and an instructions for use.
 - Ensure that the lot number of the cartridge matches that of the ID chip.
 - If the sealed cartridge has been stored in a refrigerator, place them on a clean and flat surface at room temperature for at least 30 minutes before testing.
 - Turn on the instrument for AFIAS tests.
 - Empty the tip box.
 - Insert the ID chip into the ‘ID chip port’.
- ※ Please refer to the instrument for AFIAS tests operation manual for complete information and operating instructions.

TEST PROCEDURE

► AFIAS-1, AFIAS-3, AFIAS-6

General mode

- 1) Insert a cartridge into the cartridge holder.
- 2) Insert a tip into the tip hole of the cartridge.
- 3) Select the ‘General mode’ in the instrument for AFIAS tests.
- 4) Take 100 µL of sample (whole blood/plasma/control) using a pipette and dispense it into the sample well of the cartridge.
- 5) Tap the ‘Start’ button on the screen.
- 6) The test result will be displayed on the screen after 12 minutes.

C-tip mode

- 1) Insert a cartridge into the cartridge holder.
- 2) Take 30 µL of whole blood using a C-tip.
- 3) Insert the whole blood-filled C-tip into the tip hole of the cartridge.
- 4) Select the ‘C-tip mode’ in the instrument for AFIAS tests.
- 5) Tap the ‘Start’ button on the screen.
- 6) The test result will be displayed on the screen after 12 minutes.

► AFIAS-10

Normal mode

- 1) Insert a cartridge into the cartridge holder.
- 2) Insert a tip into the tip hole of the cartridge.
- 3) Tap the ‘Load’ button of the bay that holds the cartridge with the tip to read the barcode of the cartridge and please confirm the item name written on the cartridge.
- 4) Insert the sample tube into the tube rack.
- 5) Insert the tube rack into the loading part of the sampling station.
- 6) Tap the ‘Start’ button on the screen.
- 7) The test result will be displayed on the screen after 12 minutes.

Emergency mode – General tip

- 1) The test procedure is same with the ‘Normal mode 1) – 3)’.
- 2) Convert the ‘Emergency mode’ in AFIAS-10.
- 3) Select the tip type (general tip) on the screen.
- 4) Select the sample type (whole blood/plasma) on the screen.
- 5) Take 100 µL of the sample using a pipette and dispense it into the sample well of the cartridge.
- 6) Tap the ‘Start’ button on the screen.
- 7) The test result will be displayed on the screen after 12 minutes.

Emergency mode – C-tip

- 1) Insert a cartridge into the cartridge holder.
- 2) Take 30 µL of whole blood using a C-tip.
- 3) Insert the C-tip with sample into the tip hole of the cartridge.
- 4) Tap the ‘Load’ button of the bay that holds the cartridge with a tip to read the barcode of the cartridge and please confirm the item name written on the cartridge.
- 5) Convert the ‘Emergency mode’ in AFIAS-10.
- 6) Select the tip type (C-tip) on the screen.
- 7) Tap the ‘Start’ button on the screen.
- 8) The test result will be displayed on the screen after 12 minutes.

INTERPRETATION OF TEST RESULT

- The instrument for AFIAS tests calculate the test result automatically and displays BNP concentration of the test sample in terms of pg/mL.
- Reference value: 100 pg/mL
- Working range: 5 – 5,000 pg/mL

QUALITY CONTROL

- Quality control tests are a part of the good testing practice to confirm the expected results and validity of the assay and should be performed at regular intervals.
- Quality control tests should also be performed whenever there is any question concerning the validity of the test results.
- Control materials are provided on demand with **AFIAS BNP**. For more information regarding obtaining the control materials, contact [Boditech Med Inc.’s Sales Division for assistance](#).
(Please refer to the instructions for use of control material.)

PERFORMANCE CHARACTERISTICS

- **Analytical sensitivity**
 - Limit of Blank (LoB) 2.45 pg/mL
 - Limit of Detection (LoD) 3.65 pg/mL
 - Limit of Quantitation (LoQ) 5.00 pg/mL

- **Analytical Specificity**
 - **Cross-reactivity**
Biomolecules such as below the ones in the table were added to the test sample(s) at concentrations much higher than their normal physiological levels in the blood. **AFIAS BNP** test results did not show any significant cross-reactivity with these biomolecules.

Cross-reactants	Conc.
CK-MB	1,000 ng/ml
Troponin complex	1,000 ng/ml
NT-proBNP	1,000 ng/ml
Myoglobin	200 ng/ml

- **Interference**
Interferents listed in the following table were added to the test sample at the concentration mentioned below. **AFIAS BNP** test results did not show any significant interference with these materials.

Interferents	Conc.
L-Ascorbic acid	350 µmol/L
Bilirubin	350 µmol/L
Cholesterol	13 mmol/L
D-Glucose	1,000 mg/dL
Hemoglobin	2 g/L
Triglyceride mixture	500 mg/dL
EDTA	3.4 µmol/L
Heparin	3000 U/L
Sodium citrate	2 mg/mL

■ Precision

- Single-site study

Repeatability (within-run precision)
within-laboratory precision (Total precision)
Lot to lot precision

3 Lots of **AFIAS BNP** were tested for 20 days. Each standard material was tested 2 times per day. For each test, each material was duplicated.

BNP [pg/mL]	Single-site study					
	Repeatability		within-laboratory precision		Lot to lot precision	
	AVG [pg/mL]	CV (%)	AVG [pg/mL]	CV (%)	AVG [pg/mL]	CV (%)
10	10.02	6.48	10.05	6.0	10.05	6.0
156	156.75	6.38	156.47	6.1	155.86	5.6
1250	1268.86	5.48	1266.15	5.5	1255.02	5.5

- Multi-site study

Reproducibility

1 Lot of **AFIAS BNP** was tested for 5 days in 3 different sites (1 person per 1 site, 1 instrument per 1 site). Each standard material was tested 1 time per and 5 replicates per day.

BNP [pg/mL]	Multi-site study		
	Reproducibility		
	AVG [pg/mL]	CV (%)	
10	10.01	5.6	
156	155.19	5.2	
1250	1244.76	5.6	

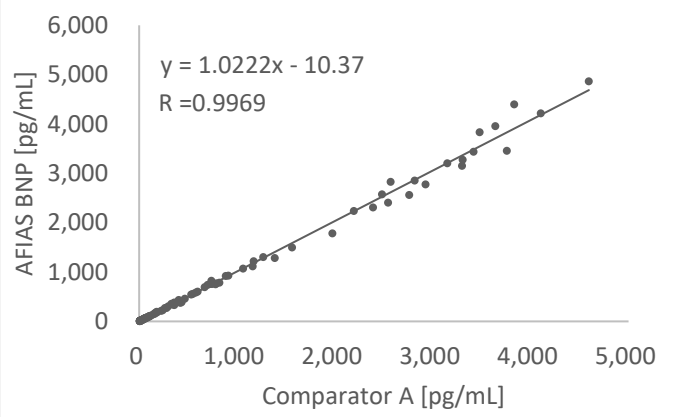
■ Accuracy

The accuracy was confirmed by testing with 3 different lots of **AFIAS BNP**. The tests were repeated 10 times at each concentration of the control standard.

BNP [pg/mL]	Lot 1	Lot 2	Lot 3	AVG [pg/mL]	Recovery (%)
10	9.99	10.32	10.08	10.13	101.3
25	23.89	25.08	25.96	24.98	99.9
50	49.10	51.62	50.19	50.31	100.6
100	102.50	102.25	99.67	101.47	101.5
250	250.96	249.39	252.12	250.82	100.3
500	501.64	476.36	489.10	489.04	97.8
1000	1014.81	993.12	1009.10	1005.68	100.6
2000	2011.20	2016.87	2001.55	2009.87	100.5
4500	4528.32	4465.50	4397.93	4463.92	99.2

■ Comparability

BNP concentration of 100 clinical samples were quantified independently with **AFIAS BNP (AFIAS-6)** and **comparator A** as per prescribed test procedures. Test results were compared, and their comparability was investigated with linear regression and correlation coefficient (R). The regression equation and correlation coefficient are as follow.



REFERENCES

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2. Mueller, T. Diagnostic accuracy of B type natriuretic peptide and amino terminal proBNP in the emergency diagnosis of heart failure. Heart, 2005, 91(5), 606–612.
3. Grantham, J. A., Borgeson, D. D., & Burnett, J. C. BNP: pathophysiological and potential therapeutic roles in acute congestive heart failure. American Journal of Physiology-Regulatory, Integrative and Comparative Physiology, 1997, 272(4), R1077–R1083.
4. Dao, Q., Krishnaswamy, P., Kazanegra, R., Harrison, A., Amirnovin, R., Lenert, L., ... Maisel, A. S. Utility of B-type natriuretic peptide in the diagnosis of congestive heart failure in an urgent-care setting. Journal of the American College of Cardiology, 2001, 37(2), 379–385

Note: Please refer to the table below to identify various symbols.

	Sufficient for <n> tests
	Read instruction for use
	Use by Date
	Batch code
	Catalog number
	Caution
	Manufacturer
	Authorized representative of the European Community
	In vitro diagnostic medical device
	Temperature limit
	Do not reuse
	This product fulfills the requirements of the Directive 98/79/EC on in vitro diagnostic medical devices

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